جامعة حفر الباطن
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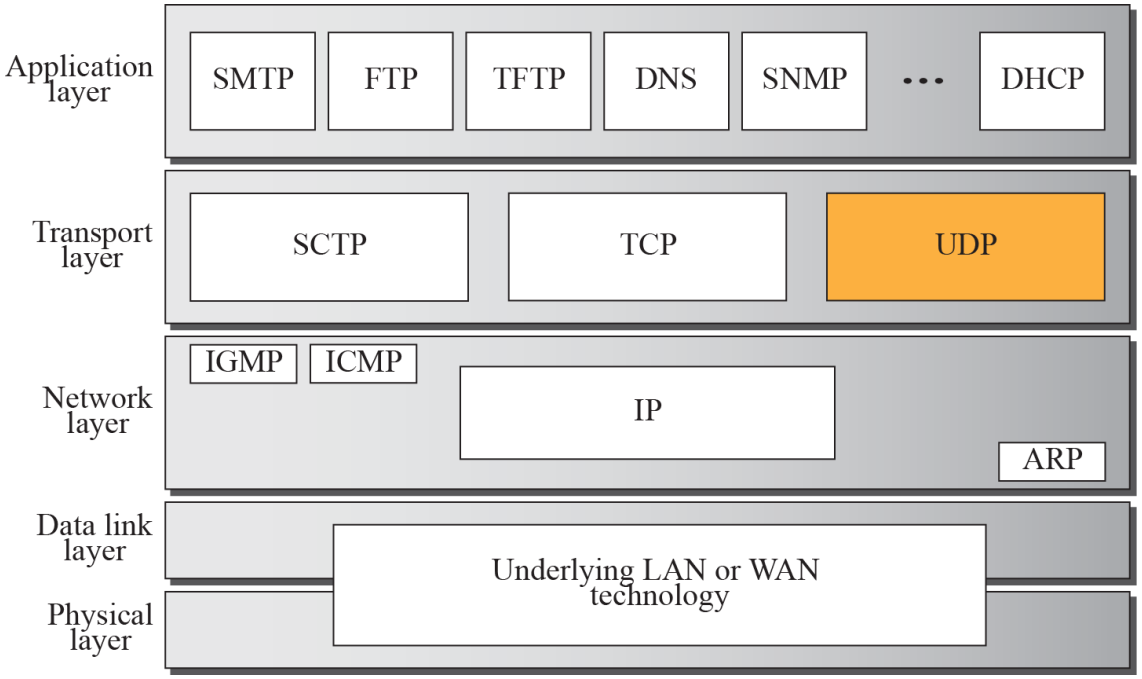
Lecture 9 - User Datagram Protocol (UDP)

Objectives

- User Datagram.
- UDP Services.
- UDP Flow Control.
- UDP Error Control.
- UDP Checksum.
- UDP Congestion Control.
- UDP Encapsulation and Decapsulation.
- UDP Features.

UDP

- UDP is located between the application layer and the network layer, and serves as the intermediary between the application programs and the network operations.



UDP

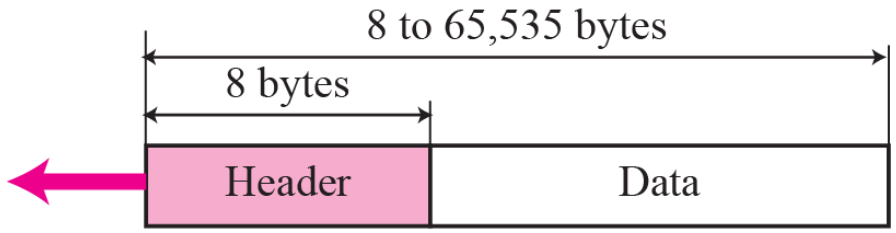
- A transport layer protocol usually has several responsibilities.
  - One is to create a process-to-process communication.
- UDP uses port numbers to accomplish this.
- Another responsibility is to provide control mechanisms at the transport level.
- UDP does this task at a very minimal level.
  - There is no flow control mechanism and there is no acknowledgment for received packets.
  - UDP, however, does provide error control to some extent. **If UDP detects an error in the received packet, it silently drops it**.

UDP

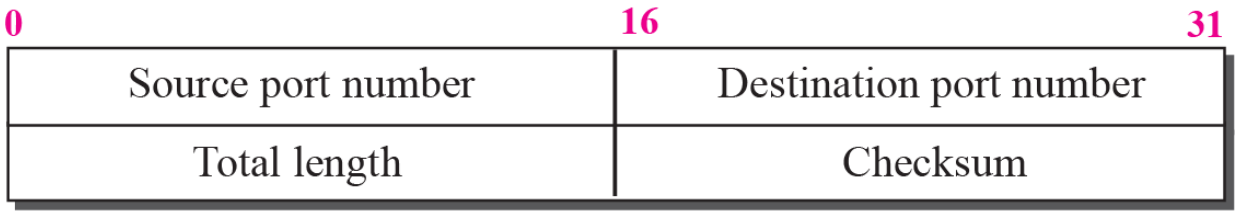
- UDP is a **connectionless, unreliable transport protocol**.
  - It does not add anything to the services of IP…
  - except for providing process-to-process communication instead of host-to-host communication.
- **If UDP is so powerless, why would a process want to use it?**
  - UDP is a very simple protocol using a minimum of overhead.
  - If a process wants to send a small message and does not care much about reliability, it can use UDP.

User Datagrams

- UDP packets, called user datagrams, have a **fixed-size header of 8 bytes**.
- Format of a user datagram *header*:
  1. **Source port number.**
  2. **Destination port number.**
  3. **Total Length.**
     - Total length of the user datagram, header plus data.
  4. **Checksum.**
     - Used to detect errors.



a. UDP user datagram



b. Header format

UDP Services

- The transport layer protocols provide general services.
- Some services provided by UDP using the following port numbers:

Port	*Protocol*	*Description*
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Domain	Domain Name Service (DNS)
67	Bootps	Server port to download bootstrap information
68	Bootpc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)

UDP Flow Control

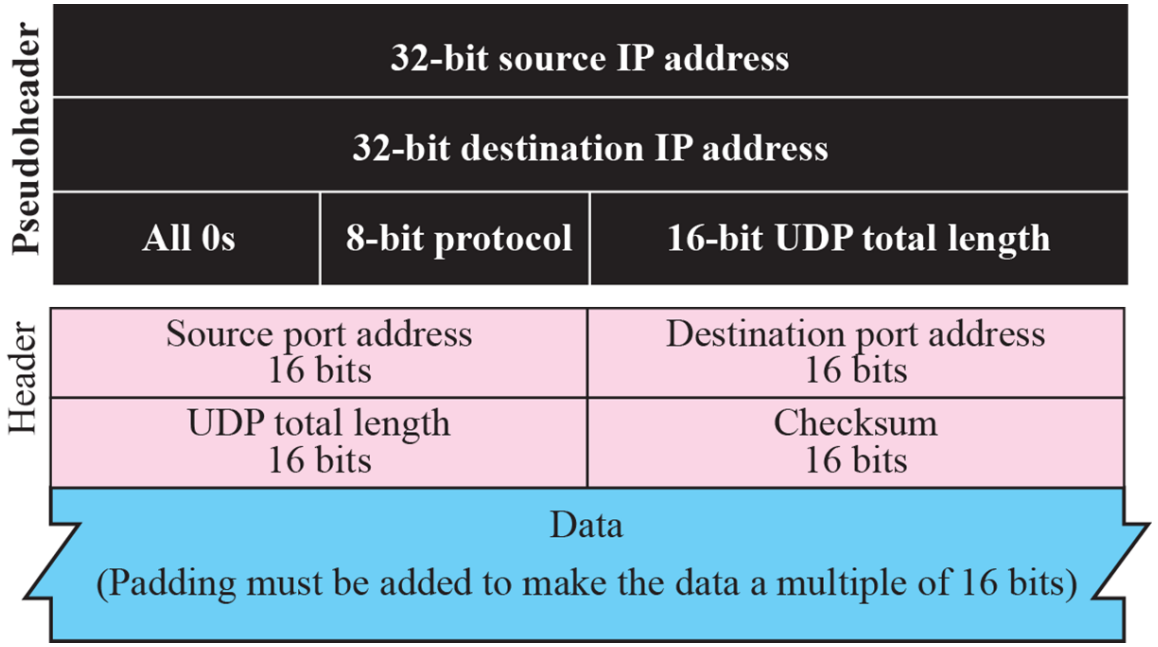
- UDP is a very simple protocol. ***There is no flow control***, and hence no window mechanism.
  - The receiver may overflow with incoming messages.
- The lack of flow control means that the process using UDP should provide for this service, if needed

UDP Error Control

- There is ***no error control mechanism in UDP except for the checksum***.
  - This means that the sender does not know if a message has been lost or duplicated.
- When the receiver detects an error through the checksum, the user datagram is silently discarded.
- The lack of error control means that the process using UDP should provide for this service if needed.

UDP Checksum

- UDP checksum calculation is different from the one for IP.
- Here the checksum includes three sections:
  - A pseudoheader.
  - The UDP header.
  - The data coming from the application layer.

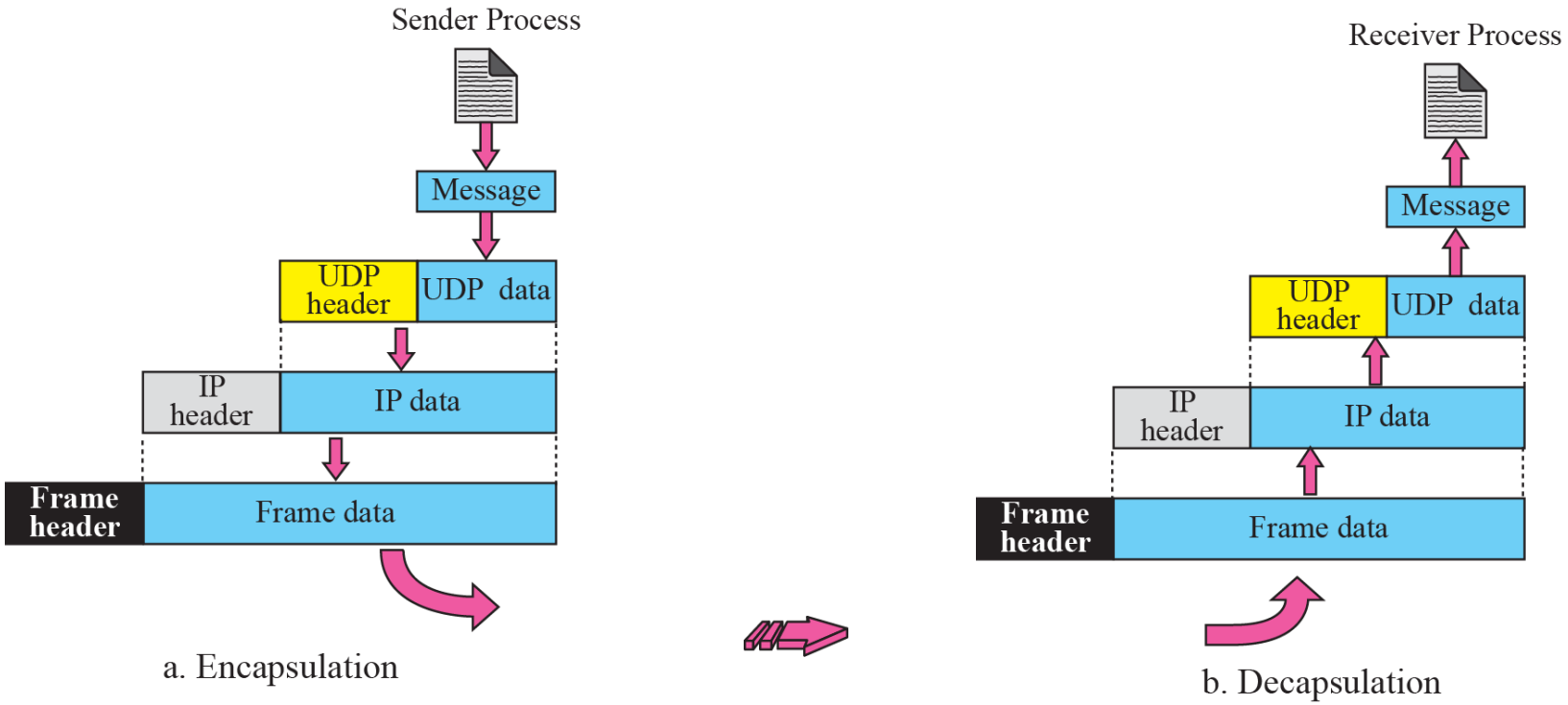


UDP Congestion Control

- Since UDP is a connectionless protocol, it **does not provide congestion control**.
- UDP *assumes* that the packets sent are small and infrequent, and cannot create congestion in the network.
- **This assumption may or may not be true today when UDP is used for real-time transfer of audio and video.**

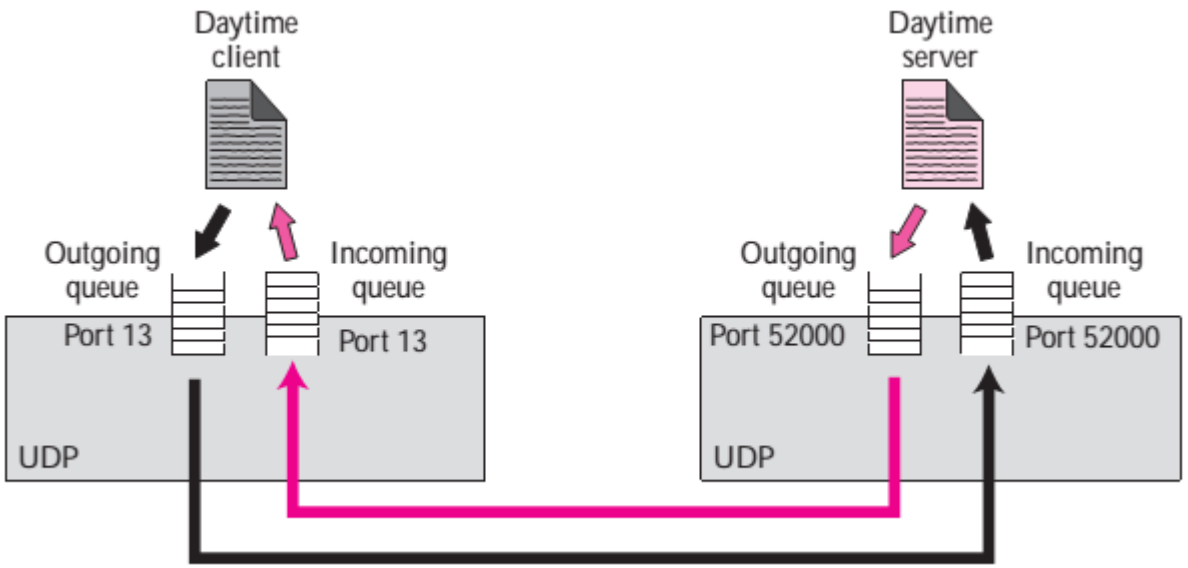
UDP Encapsulation and Decapsulation

- To send a message from one process to another, the UDP protocol encapsulates and decapsulates messages.



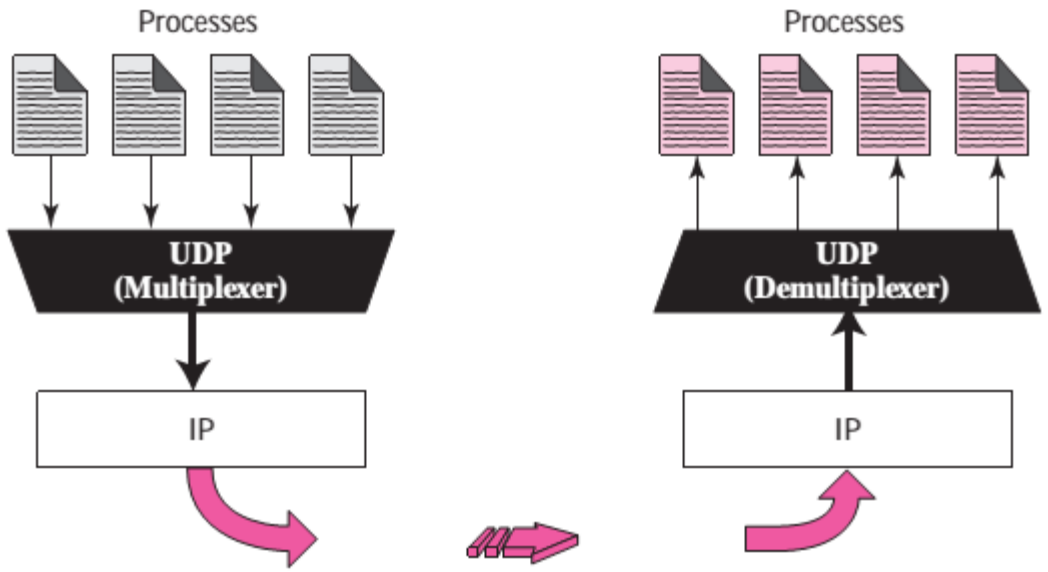
UDP Queuing

- At the client site, when a process starts, it requests a port number from the operating system.
- Some implementations create both an incoming and an outgoing queue associated with each process.



UDP Multiplexing and Demultiplexing

- In a host running a TCP/IP protocol suite, there is only one UDP but possibly several processes that may want to use the services of UDP.
- To handle this situation, *UDP multiplexes and demultiplexes*.



UDP Features

1. Connectionless Protocol:

- UDP is a connectionless protocol. Each UDP packet is independent from other packets sent by the same application program
- Example:
  - DNS uses the services of UDP because a client needs to send a short request to a server and to receive a quick response from it.
  - The request and response can each fit in one user datagram.
  - only one message is exchanged in each direction, out-of-order delivery is not an issue.

UDP Features

2. Lack of Congestion Control:

- UDP does not provide congestion control.
- However, UDP does not create additional traffic in an error-prone network.
  - TCP may resend a packet several times and thus contribute to the creation of congestion or worsen a congested situation.
- Therefore, in some cases, lack of error control in UDP can be considered an advantage when congestion is a big issue.

UDP Applications

- UDP is more suitable than TCP in the following typical applications:

1. UDP is a suitable for a process with a **simple request-response communication** with little concern for flow and error control.
2. UDP is a suitable for a process with **internal flow and error-control mechanisms**.
   - For example, the Trivial File Transfer Protocol (TFTP) process includes flow and error control. It can easily use UDP.
3. UDP is normally used **for real-time applications**
   - Applications that cannot tolerate uneven delay between sections of a received message.
4. UDP is a suitable transport protocol **for multicasting.**
5. UDP is used **for some route updating protocols** (such as RIP).
6. UDP is used **for management processes** such as SNMP